

Quiz (Habanero)

- Q1.** A water tank can hold 480 gallons of water. If 3 sprinklers use 15 gallons each per day, how many days will the tank last? Use divisibility rules to find the quotient.
(A) 8
(B) 9
(C) 10
(D) 12
(E) 15
- Q2.** What is the remainder when 53 is divided by 5?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q3.** A park ranger recorded 27 degrees Celsius at 8 AM and 37 degrees at 2 PM. What is the difference in temperature? If the difference is divisible by 3, what is the quotient?
(A) 8
(B) 9
(C) 10
(D) 11
(E) 12
- Q4.** In an ecosystem, a tree absorbs 18 gallons of water per day. If the tree needs to absorb 90 gallons of water to reach its daily limit, how many days will it take? Use division with remainder to find the answer.
(A) 4 R 6
(B) 5 R 0
(C) 4 R 10
(D) 5 R 2
- (E) 4 R 2
- Q5.** What is the result of $24 \div 3$ using the division algorithm?
(A) 8 R 0
(B) 7 R 1
(C) 6 R 2
(D) 5 R 3
(E) 4 R 4
- Q6.** If 15 baskets can hold 45 apples, and 3 apples are taken from each basket, how many apples are left in each basket? Use the divisibility rule of 3 to find the answer.
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q7.** A weather station recorded 36 inches of rainfall in a month. If the rainfall was evenly distributed throughout 4 weeks, how many inches of rainfall occurred per week? Use division with remainder to find the answer.
(A) 9 R 0
(B) 8 R 2
(C) 9 R 1
(D) 8 R 4
(E) 7 R 3
- Q8.** What is the remainder when 75 is divided by 9?
(A) 3
(B) 6
(C) 0
(D) 5
(E) 4

Open Ended Questions (FRQ)

- Q9.** In a greenhouse, 48 plants are arranged in rows. If each row can hold 8 plants, how many rows can be made? Use long division to find the quotient and remainder, and explain your reasoning in the space provided below.
- Q10.** A researcher studied the temperature changes in a region and found that the average temperature increased by 18 degrees Fahrenheit over 6 years. If the temperature increase is evenly distributed throughout the years, what is the temperature increase per year? Use division with remainder to find the answer and show your work in the space provided below.

Chef's Tips

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- Tip 1: Encourage students to use visual aids to solve division problems.
- Tip 2: Remind students to check their remainder when using the division algorithm.
- Tip 3: Emphasize the importance of explaining the reasoning behind their answers.